

A counterexample to normalized poset reconstruction

Standalone research note • David Bridges • 1 October 2026

Result. Two nonisomorphic posets, each with 16 elements, have exactly the same labeled normalized principal-upset intersection matrix. This note records the construction and proof. It makes no novelty, minimality or publication claim.

1. What is being compared?

For a finite poset P , its principal upset is $U(x)=\{y : x\leq y\}$, including x . Put $K(x,y)=|U(x)\cap U(y)|$ and $d(x)=|U(x)|$. The normalized matrix is $G(x,y)=K(x,y)/\sqrt{(d(x)d(y))}$. A genuine collision means identical normalized data from posets that cannot be made identical by relabeling.

2. The attachment map

Take two disjoint copies E and W of the subsets of $\{0,1,2\}$. Each copy has eight labels. Define the map ϕ from E to W by the table below.

Subset X	$\phi(X)$
$\emptyset$	$\emptyset$
$\{0\}, \{1\}, \{2\}$	$\{0\}$
$\{0,1\}, \{0,2\}$	$\{0,1\}$
$\{1,2\}$	$\{0,2\}$
$\{0,1,2\}$	$\{0,1,2\}$

ϕ preserves inclusion and subset cardinality. Every singleton maps to $\{0\}$, and each pair maps to a pair containing 0. Write $e(X)=2^{(3-|X|)}$, the upset size inside a Boolean sector. Then $e(\phi(X))=e(X)$.

3. Define the two orders

Source P : use inclusion within both sectors, and $E_X\leq W_Y$ exactly when $\phi(X)\subseteq Y$. Candidate Q : use reverse inclusion within both sectors, and $E_X\leq W_Y$ exactly when $Y\subseteq\phi(X)$. Neither order has relations from W back to E .

Both are posets. Reflexivity and antisymmetry follow from the sector orders and one-way cross relations. Transitivity across sectors follows from inclusion preservation of ϕ . These are complete standalone examples; no embedding into a larger construction is assumed.

4. Exact equality of normalized matrices

For subsets A, B , let $k(A, B) = 2^{3 - |A \cup B|}$ and $\ell(A, B) = 2^{|A \cap B|}$. They count common successors and common predecessors in a Boolean sector, respectively. The cardinality identity $|A \cup B| + |A \cap B| = |A| + |B|$ gives:

$$e(A)e(B)\ell(A, B) = 8k(A, B).$$

Label pair	Source intersection K_P	Candidate intersection K_Q
E_X, E_Y	$k(X, Y) + k(\phi(X), \phi(Y))$	$\ell(X, Y) + \ell(\phi(X), \phi(Y))$
E_X, W_Y	$k(\phi(X), Y)$	$\ell(\phi(X), Y)$
W_X, W_Y	$k(X, Y)$	$\ell(X, Y)$

For a label i in either sector, let e_i be its sector upset size. Since ϕ preserves these sizes, every entry obeys $K_Q(i, j) = 8K_P(i, j)/(e_i e_j)$. Set $S(i, i) = \sqrt{8/e_i}$. Thus $K_Q = S K_P S$, including the diagonals.

The full diagonals are $d_P(E_X) = 2e(X)$, $d_P(W_X) = e(X)$, $d_Q(E_X) = 16/e(X)$, and $d_Q(W_X) = 8/e(X)$. These are the full-poset sizes used in normalization; $e(X)$ alone is a sector size.

Consequently $d_Q(i) = S(i, i)^2 d_P(i)$, and normalizing cancels both positive scale factors: $G_Q(i, j) = G_P(i, j)$ for every pair of labels.

5. The posets are not isomorphic

An order isomorphism preserves every element's upset and downset sizes. For the four elements with upset size two, the downset-size multisets are: $P: \{5, 8, 9, 10\}$; $Q: \{6, 7, 8, 11\}$. Therefore no relabeling can identify the orders.

In particular, $W_{\{0, 2\}}$ in P has two successors and nine predecessors: four from W and five from E . Q has no element with that pair of sizes. This is a complete nonisomorphism certificate.

6. Verification and scope

The frozen v126 construction and v127 audit include integer-matrix and independent set-based checks of all 256 entries, followed by cover-closure reconstruction and exact rational checks. An initial certificate label $W_{\{0, 1\}}$ was corrected to $W_{\{0, 2\}}$; the failed assertion and correction remain in the historical record.

Consequence: the universal claim that G determines every finite poset up to isomorphism is false. Earlier conditional results retain their hypotheses. The smallest collision size and literature novelty are unresolved. This document is an internal standalone note, not a preprint.